

Burden-Constrained Policy-Layer Authorization for Simulated EEG-Guided Adaptive Vagus Nerve Stimulation

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Abstract

EEG-guided adaptive neuromodulation requires explicit rules for converting model scores into intervention authorization. We evaluated a simulated vagus nerve stimulation (VNS) authorization framework that combines probability calibration, temporal persistence, and attribution overlap, with burden and event timing treated as policy-level outcomes rather than inferred from discrimination alone.

Five CHB-MIT cases contributed 175 EDF recordings (172.83 h), 310,916 four-second windows, and 27 annotated seizures. Model development, calibrator fitting, and threshold selection used the original 70-EDF analysis inventory; the fold-specific fitted pipelines were then evaluated over the complete 175-EDF inventory, with the other 105 recordings entering evaluation only and no operating parameter retuned on them. Authorization burden was strongly case dependent: four held-out cases produced 0.049–0.340 false authorization clusters per recording-hour, whereas chb05 produced 17.46/h, yielding 700 false clusters overall and a pooled rate of 4.05/h. Complete-recording evaluation therefore exposed a pronounced case-specific generalization failure despite satisfaction of the training-side burden constraint. In a common-score factorial analysis, temporal persistence accounted for most of the reduction in authorization burden, whereas attribution overlap provided a smaller incremental restriction. At the earliest complete-window decision time, 2/27 seizures were authorized before onset, 18/27 at or after onset, and 7/27 were missed; median latency among covered events was 16 s (interquartile range 8.5–28 s). Reclassifying false-cluster truth at window-end decision availability left the principal burden result unchanged. Adapted majority-voting and firing-power comparators occupied similar burden–coverage ranges under a common 300 s refractory rule.

These findings show that discrimination and calibration-side operating constraints alone do not characterize an adaptive authorization policy. Complete recording exposure, temporal post-processing, causal decision-time availability, attribution constraints, and case-level burden must be evaluated explicitly. The framework provides a reproducible policy-layer methodology for simulated EEG-guided neuromodulation while distinguishing seizure-responsive authorization from advance prediction and therapeutic validation.

Keywords: explainable AI; EEG; adaptive neuromodulation; vagus nerve stimulation; calibration; authorization policy; attribution overlap.

1 Introduction

Epilepsy affects more than 50 million people worldwide, with more than 4.9 million new cases each year, and approximately one-third of people with epilepsy remain drug resistant despite

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appropriately selected antiseizure medications[7]. For patients who remain refractory and are not candidates for curative resective surgery, neuromodulation is an important therapeutic option. Vagus nerve stimulation is used as adjunctive therapy for selected patients with refractory epilepsy, with programming shaped by dose, duty cycle, tolerability, adverse effects, battery use, and longitudinal clinical judgment[4, 10].

In adaptive neuromodulation, the relevant decision is not simply whether stimulation is delivered according to a fixed schedule, but how sensed physiological activity is used to trigger, withhold, or modify stimulation in real time[11, 15]. Contemporary VNS already implements one form of biomarker-triggered stimulation through ictal tachycardia detection[5]. EEG-derived estimates offer a more direct route for seizure-responsive triggering, but they also expand the decision chain: intervention authorization depends on signal acquisition, preprocessing, feature extraction, label definition, model structure, calibration, thresholding, temporal post-processing, and the rule used to consolidate repeated outputs into operational episodes[2, 16].

This expanded decision chain is especially important in retrospective EEG simulation, where the recorded seizure interval is observed but the preictal state is inferred by convention. In CHB-MIT, seizure annotations define observed ictal start and stop times in pediatric scalp EEG recordings made available through PhysioNet[6, 8]. By contrast, preictal labels are modeling constructs imposed around those intervals, and seizure-prediction studies emphasize that preictal duration, alarm generation, and post-processing choices materially affect operational performance[1, 2]. Consequently, choices of preictal horizon, window length, feature representation, validation scheme, thresholding, and post-processing define different operational risk scores and alarm or authorization policies.

Although AUROC remains useful for characterizing discrimination, it is insufficient for systems that act because it summarizes ranking performance rather than calibrated risk, threshold-specific utility, or operational consequences[3, 13]. This limitation is particularly important for seizure prediction and detection, where sample-level performance can diverge markedly from alarm-level behavior once temporal regularization, refractory intervals, false alarms per hour, and event timing are introduced[1, 2]. Deployment-oriented work has likewise emphasized false-alarm burden and temporal behavior in wearable EEG seizure detection, including artifact filtering and boosted-tree models evaluated on CHB-MIT[9].

Explainability addresses a different part of the decision chain. SHAP and related attribution methods can identify which EEG features or channels contribute to seizure-related predictions, and recent work has used XAI to reduce the feature and channel set required for seizure detection[14]. Such use is primarily interpretive or model-reduction oriented. If attribution information is used inside an authorization policy, a separate question arises: whether the attribution structure is temporally stable enough to serve as a reproducible constraint, rather than whether it is clinically meaningful in its own right. That distinction is important because explanation reliability and uncertainty are not interchangeable[12].

Recent related studies therefore address complementary, but non-equivalent, stages of the pipeline (Table 1; Figure 1). Andrade et al. evaluated sample- and alarm-based seizure-prediction performance using firing-power post-processing and a refractory period[1]. Batista et al. examined post-processing as an explicit chronology of seizure-prediction decisions[2]. Ingolfsson et al. emphasized artifact-aware false-alarm reduction and wearable implementation constraints[9]. Vieira et al. used XAI for feature and channel reduction in seizure detection[14]. The published performance numbers are not directly exchangeable because cohorts, targets, validation schemes, and alarm definitions differ. Alongside the methodological comparison, we therefore implement adapted temporal voting and firing-power mechanisms on common calibrated scores, training-only parameter selection, and identical held-out recordings. This tests decision components under controlled conditions without claiming replication of the external systems in their entirety. Unlike device-oriented work, the present study delivers no stimulation and evaluates neither artifact suppression nor embedded hardware; its scope is retrospective

authorization analysis in five cases.

Table 1: Methodological comparison with recent EEG seizure-prediction and seizure-detection studies. “Not reported” indicates that the cited study did not present the item as a formal component of the evaluated pipeline; it does not imply that the item is unimportant to that application.

Study	Primary task	Temporal or operational rule	Calibration and burden reporting	Attribution/XAI role
Andrade et al. (2024)[1]	Seizure prediction across databases	Firing-power regularization with a refractory period	Probability calibration not reported; sensitivity and false predictions/h reported	Not part of the decision rule
Batista et al. (2024)[2]	Patient-specific seizure prediction	Explicit post-processing chronology around preictal alarms	Probability calibration not reported; alarm-level consequences evaluated	Not part of the decision rule
Ingolfsson et al. (2024)[9]	Seizure detection for wearable EEG	Artifact detector combined with a boosted-tree seizure detector	Probability calibration not reported; false positives/h and sensitivity emphasized	Not part of the decision rule
Vieira et al. (2023)[14]	Seizure detection	Window-level classification	Probability calibration and cluster burden not reported	SHAP used for feature and channel interpretation and reduction
Present study	Simulated EEG-guided VNS authorization	Train-only threshold selection, temporal persistence, attribution-overlap constraint, and cluster/cooldown rules	Nested calibration; false authorization clusters per analyzed recording-hour; event-level timing	Top- k attribution overlap used as a policy constraint and evaluated across temporal lags

Andrade et al. (2024)	●	–	●	●	–	–
Batista et al. (2024)	●	–	●	●	–	–
Ingolfsson et al. (2024)	●	–	●	●	–	–
Vieira et al. (2023)	●	–	–	–	●	–
Present study	●	●	●	●	●	●
	Score / model	Probability calibration	Temporal rule	Burden / timing	Attribution / XAI	Simulated VNS authorization

Figure 1: Methodological coverage of representative related work. Filled cells indicate that a stage was explicitly modeled or evaluated in the cited pipeline. The comparison is structural rather than a ranking of predictive performance; the last column refers specifically to simulated VNS authorization.

By separating score estimation from the rules that permit simulated intervention, this study tests how probability calibration, temporal persistence, and attribution overlap affect authorization burden and event timing across heterogeneous EEG recordings. Because the operational target combines preictal and ictal windows, the framework is evaluated as a seizure-responsive authorization system rather than assumed to be an advance seizure-prediction system. Pre-onset authorization is therefore treated as a separate timing outcome, allowing predictive and post-

onset behavior to be distinguished explicitly. Attribution overlap is treated as an operational policy variable rather than as a surrogate for clinical explanation validity.

2 Methods

2.1 Framework

The framework was implemented as a modular Python workflow comprising data access, windowing, feature extraction, model fitting, probability calibration, feature attribution, authorization-policy simulation, event-level timing, and burden estimation. Figure 2 summarizes the analysis chain. All reported VNS outcomes are simulated authorizations; no stimulation was delivered.

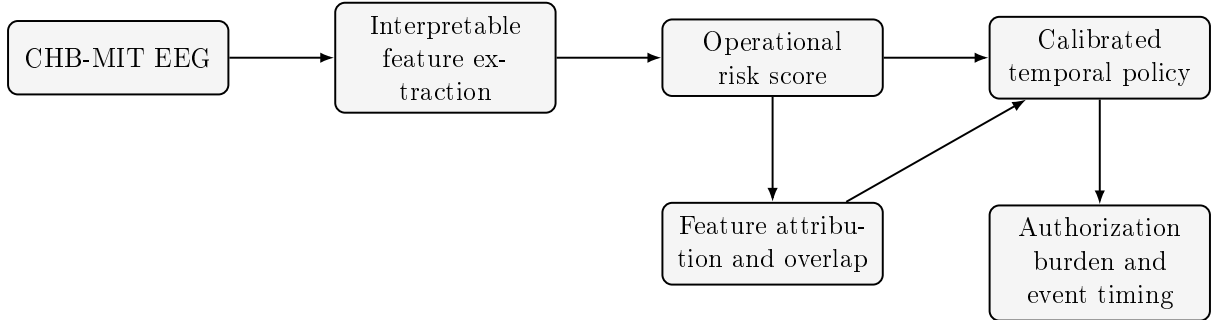


Figure 2: Simulated EEG-guided authorization framework. EEG windows are converted into features and operational risk scores; calibration, temporal persistence, and attribution-overlap constraints determine authorization; burden and event timing are evaluated separately from model discrimination.

2.2 Dataset, study sample, and labels

The source was the CHB-MIT scalp EEG database, version 1.0.0[8]. We analyzed five cases (chb01, chb02, chb03, chb05, and chb08), comprising all 175 EDF recordings available for those cases, 172.8294 h of EEG, and 27 annotated seizures. These cases form a convenience sample; no documented prospective clinical sampling criterion is available for their selection, so they are not treated as a representative validation cohort.

The original 70-EDF analysis inventory (67.8214 h) comprised all 27 seizure-bearing EDFs and 43 EDFs without annotated seizures. Within each fold, only the designated core and calibration partitions from this inventory contributed to model fitting, calibrator fitting, or threshold selection. The other 105 EDFs (105.0081 h), all without seizure annotations, entered only the expanded held-out evaluation. All 175 files matched the SHA-256 values in the PhysioNet version-1.0.0 registry and passed EDF header/file-length and 22-channel, 256-Hz montage checks. Table 2 summarizes the recording inventory.

Windows were 4 s long with a 2 s step. Window-center labels were ictal inside the annotated onset–offset interval, preictal during the preceding 300 s, and operationally negative otherwise. The binary target combined preictal and ictal labels. Accordingly, the binary target represents seizure-responsive authorization over the defined preictal-to-ictal interval; pre-onset performance is evaluated separately at the event level rather than being implied by the positive class definition. Labeling and temporal rules reset at EDF boundaries; preictal labeling was not propagated into a preceding EDF. Consequently, recordings without an annotated seizure contain no positive operational labels under this record-local convention, which does not imply that every sample is physiologically interictal. Across the development subset there were 5,227 positive windows.

Table 2: EEG recording inventory and analysis role. The original analysis-inventory EDFs supply the fold-specific core, calibration, and development-test partitions; additional evaluation-only EDFs contribute only to held-out scoring.

Case	Original inventory EDFs	Additional evaluation-only EDFs	Total EDFs	Total h	Seizures
chb01	20	22	42	40.55	7
chb02	9	27	36	35.27	3
chb03	13	25	38	38.00	7
chb05	14	25	39	39.00	5
chb08	14	6	20	20.01	5
Total	70	105	175	172.83	27

2.3 Validation design and nested calibration partitions

The 70 model-development EDFs were analyzed with held-out-subject validation and within-subject chronological EDF splits using a 0.40 outer test fraction. For nested calibration, the outer-training recordings were divided into a model-fitting core and a calibration block. Calibration used the last 30% of outer-training recordings within each subject, with a predefined class-preserving fallback when either block otherwise lacked one class. Core, calibration, and development-test EDFs were disjoint, and overlapping windows never crossed partition boundaries. Exact EDF identities and per-partition hours, windows, positive windows, and seizure-containing recordings are provided in the supplementary tables and software.

For the principal held-out-subject analysis, each fitted model and its calibrator were evaluated across the full EDF inventory of the held-out case. The 105 test-only recordings were never introduced into training or parameter selection. This design tests recording-level generalization within the same five cases; it is not independent external validation in new patients.

2.4 Features and predictive models

The 20-feature vector comprised absolute and relative power in delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–50 Hz) bands; spectral entropy; Hjorth activity, mobility, and complexity; line length; root-mean-square amplitude; variance; zero-crossing rate; skewness; and kurtosis. Each feature was averaged over the 22 analysis channels. Welch spectra used 256-sample segments. EEG samples were represented in volts as float32 values; duplicate channel labels were resolved deterministically, and nonfinite feature values were replaced by zero. The same feature pipeline was applied to every EDF.

Logistic regression, random forest, and LightGBM were used to characterize the score generator. Thresholded model-comparison metrics used $p \geq 0.5$. Policy analyses used nested-core LightGBM fits with 100 trees, learning rate 0.1, 31 maximum leaves, unrestricted depth, 20 minimum child samples, balanced class weights, and random seed 42. Because calibration recordings are withheld from nested-core fitting, nested LightGBM discrimination differs slightly from the complete outer-training model-comparison estimates.

2.5 Probability calibration and train-only operating-point selection

Platt scaling and isotonic regression were fitted only on nested calibration recordings after LightGBM fitting. Calibration was summarized using Brier score, expected calibration error (ECE), reliability bins, calibration slope/intercept, and score range.

For the primary factorial analysis, the calibrator and probability threshold were selected using training data only. Sigmoid and isotonic calibration were compared by leave-one-EDF-out cross-fitted Brier score within the calibration block; the lower-Brier mapping was refit on

the complete calibration block. Thresholds from 0.01 to 0.99 were evaluated on cross-fitted calibration predictions. Thresholds producing no more than 2 false authorization clusters per calibration recording-hour were considered feasible; among them, selection maximized positive-window sensitivity, then preferred lower false-cluster burden and lower threshold. Temporal persistence was fixed at two consecutive above-threshold windows and the attribution-overlap threshold at $\gamma = 0.4$.

2.6 Attribution overlap

LightGBM per-window contributions were obtained with `pred_contrib=True`; the bias column was removed and absolute contributions were ranked over the fixed 20-feature order. For lag ℓ , the top-five attribution overlap was

$$S_{t,\ell} = \frac{|T_5(E_t) \cap T_5(E_{t-\ell})|}{|T_5(E_t) \cup T_5(E_{t-\ell})|}.$$

Only pairs from the same EDF with the required separation were valid. The policy used adjacent-window overlap ($\ell = 1$) and required $S_{t,1} > 0.4$, corresponding to at least three shared top-five features. Sensitivity analyses evaluated 2, 4, 6, 10, and 20 s center separations; separations of at least 4 s contain no shared raw EEG samples. An outcome not used to define the gate was the within-case difference in mean overlap between positive-label and negative-label authorized windows. Attribution overlap is therefore evaluated as a ranked-model-contribution property, not as clinical explanation ground truth.

2.7 Authorization policies and burden endpoints

The primary factorial comparison used one calibrated probability scale and one train-selected threshold within each fold. Four configurations independently toggled temporal persistence and the $S_t > 0.4$ attribution-overlap constraint: calibrated threshold only; threshold plus persistence; threshold plus attribution overlap; and threshold plus both. This isolates the incremental contribution of the two policy constraints conditional on the calibrated score and selected operating point.

Authorization masks were computed in EDF-local timestamp order. Selected windows were merged into a cluster when the next selected window started no later than the current cluster end plus the stated cooldown; EDF boundaries always terminated clusters. A cluster was false only when none of its selected windows carried a positive operational label. A cluster crossing a label boundary was therefore non-false if at least one selected constituent window was positive. Cooldown changes cluster consolidation without changing the authorization mask and therefore preserves a policy’s own event coverage. Refractory suppression, used only in the comparator benchmark, removes subsequent decisions within the refractory interval and can change coverage.

The primary burden denominator is the sum of complete durations of unique evaluated EDFs, including labeled preictal and ictal time. We therefore report false authorization clusters per analyzed recording-hour rather than an interictal-only false-alarm rate. We also report selected-window fraction, positive authorization rate (positive-label selected windows divided by all selected windows), and selected-support duration where relevant. Equal-weight case means and pooled recording-hour rates are reported separately.

2.8 Event mapping and decision-time availability

For a seizure with onset a and offset b , event coverage required at least one authorization timestamp in $[\max(0, a - 300), b]$. A first authorization before a was classified as pre-onset; a first authorization at or after a as at-or-after-onset only; and an event without authorization as missed. Latency is first authorization time minus onset, and lead is its negative.

Because a complete 4 s feature window is not available at its center, the principal event analysis uses the window end as the earliest causal decision timestamp. No unmeasured computation, transmission, clinician, or device delay was added. The principal timing summary is the pooled median latency and interquartile range among covered seizures.

2.9 Common-score temporal comparators

Temporal decision mechanisms were compared on identical model outputs. Majority-of-three voting and a firing-power rule were applied to the same calibrated LightGBM scores used by the threshold, persistence, and persistence-plus-overlap policies. Majority voting accepted at least two of the current and two preceding binary threshold decisions. Firing power averaged the last m binary decisions and accepted when the fraction exceeded 0.7, with $m \in \{5, 15, 30, 60, 150\}$. These are component-level adaptations of temporal mechanisms used in recent seizure-prediction/detection work[1, 2, 9], not reproductions of those studies' complete classifiers, cohorts, or hardware.

For each family, thresholds and history length were selected using calibration data only. Feasible candidates met the same 2 false-clusters/h calibration constraint; selection then maximized window-end event coverage, minimized false-cluster burden, maximized pre-onset count, and preferred lower thresholds/shorter histories. The benchmark was evaluated both without refractory suppression and with a common 300 s refractory rule. Main-text comparisons use the common 300 s rule; the unsuppressed results are supplied in the supplement because prolonged authorization can collapse into few clusters and thereby make cluster frequency alone misleading.

2.10 Statistical summaries and reproducibility

Rates and case-level coverage are reported as equal-weight means $\pm$ sample SD across the five held-out cases. Pooled recording-hour rates divide summed counts by summed recording duration; pooled event coverage divides covered events by all annotated seizures. Exploratory 95% percentile intervals enumerate all $5^5 = 3125$ ordered case-bootstrap draws, with paired configurations resampled jointly. Descriptive two-sided sign-flip probabilities enumerate all $2^5 = 32$ case-level sign patterns. With five cases and overlapping LOSO training sets, these summaries are descriptive rather than confirmatory population inference.

Code, fitted LightGBM boosters and calibrators, feature order, derived inputs, per-window scores, attribution ranks, cluster/event ledgers, parameter searches, figure-generation scripts, and tests are included in the versioned reproducibility package. Raw CHB-MIT EEG is not redistributed.

2.11 Targeted decision-time and case diagnostics

Two targeted robustness analyses were performed after completion of the main revision analyses, without refitting models or retuning any operating parameter. First, the selected authorization masks and support-defined clusters were held fixed while cluster truth was recomputed from operational labels evaluated at window-end decision availability rather than window centers. This tests whether the false-cluster endpoint is sensitive to the same causal timing convention used for event-level authorization. Second, the chb05 burden was decomposed by EDF for the train-selected persistence-plus-overlap policy. We quantified the number and share of false clusters contributed by each recording, compared the original analysis-inventory and additional evaluation-only exposures, and summarized the upper tail of calibrated scores among operationally negative windows. The latter comparison is descriptive and is used to characterize where the held-out burden arises, not to define a new operating point.

3 Results

3.1 Score-generator performance and calibration

Held-out-subject model discrimination on the development partitions was moderate (Table 3). LightGBM AUROC was 0.620 ± 0.098 and AUPRC 0.142 ± 0.062 ; the corresponding within-subject values were 0.679 ± 0.231 and 0.269 ± 0.203 . These results motivate treating the classifier as an operational score generator rather than the principal endpoint.

Table 3: Development-partition window-level discrimination. Values are mean $\pm$ SD across five folds; thresholded metrics use $p \geq 0.5$.

Validation	Model	AUROC	AUPRC	Sens.	Spec.	F1	Bal. acc.
Held-out	Logistic regression	0.551 ± 0.114	0.159 ± 0.115	0.358 ± 0.167	0.764 ± 0.160	0.114 ± 0.053	0.561 ± 0.063
	Random forest	0.597 ± 0.037	0.124 ± 0.064	0.058 ± 0.046	0.995 ± 0.010	0.095 ± 0.076	0.527 ± 0.022
	LightGBM	0.620 ± 0.098	0.142 ± 0.062	0.328 ± 0.145	0.853 ± 0.064	0.137 ± 0.041	0.591 ± 0.048
Within-subject	Logistic regression	0.659 ± 0.240	0.197 ± 0.150	0.572 ± 0.245	0.676 ± 0.137	0.135 ± 0.081	0.624 ± 0.185
	Random forest	0.660 ± 0.229	0.267 ± 0.204	0.174 ± 0.147	0.996 ± 0.002	0.258 ± 0.204	0.585 ± 0.074
	LightGBM	0.679 ± 0.231	0.269 ± 0.203	0.295 ± 0.224	0.928 ± 0.023	0.195 ± 0.154	0.611 ± 0.121

The nested-core LightGBM score evaluated across the full 175-EDF held-out inventory had raw AUROC 0.601 ± 0.093 and AUPRC 0.082 ± 0.059 (Table 4). The isotonic mapping selected on calibration data reduced mean Brier score from 0.133 to 0.024 and ECE from 0.239 to 0.024. Sigmoid calibration had a slightly lower full-inventory Brier score, but it was not substituted post hoc because isotonic calibration had been selected from cross-fitted calibration data.

Table 4: Nested-core LightGBM score diagnostics over the full held-out recording inventory. Values are equal-weight case means $\pm$ SD.

Mapping	AUROC	AUPRC	Brier	ECE
Raw	0.601 ± 0.093	0.082 ± 0.059	0.133 ± 0.067	0.239 ± 0.101
Sigmoid	0.601 ± 0.093	0.082 ± 0.059	0.019 ± 0.009	0.020 ± 0.015
Isotonic	0.590 ± 0.031	0.073 ± 0.051	0.024 ± 0.016	0.024 ± 0.025

3.2 Train-selected operating points

Cross-fitted Brier score selected isotonic calibration in all five held-out folds. Train-selected probability thresholds ranged from 0.22 to 0.48, and each calibration fold satisfied the predefined ≤ 2 false-clusters/recording-hour constraint before held-out evaluation (Table 5).

Table 5: Training-only calibrator and threshold selection for held-out-subject analysis.

Held-out case	Calibrator	Brier: Platt	Brier: isotonic	θ	Cal. false clusters/h
chb01	Isotonic	0.0362	0.0319	0.24	1.87
chb02	Isotonic	0.0327	0.0296	0.44	1.49
chb03	Isotonic	0.0265	0.0248	0.34	1.64
chb05	Isotonic	0.0323	0.0299	0.48	1.20
chb08	Isotonic	0.0319	0.0294	0.22	1.90

3.3 Held-out authorization burden across the full recording inventory

Authorization burden varied markedly across cases (Table 6; Figure 3). The persistence-plus-overlap policy produced 700 false clusters over 172.83 h, a pooled rate of 4.050/h. The equal-weight case mean was 3.601 ± 7.748 /h, whereas the case median was 0.105/h. Four cases ranged

from 0.049 to 0.340/h, while chb05 produced 681 false clusters (17.460/h). Thus, the 2/h constraint used during calibration did not generalize as a subject-independent held-out bound.

Table 6: Case-level false-cluster burden for the train-selected persistence-plus-attribution-overlap policy over all 175 EDFs.

Case	False clusters/h	False clusters	Evaluated h
chb01	0.049	2	40.55
chb02	0.340	12	35.27
chb03	0.105	4	38.00
chb05	17.460	681	39.00
chb08	0.050	1	20.01

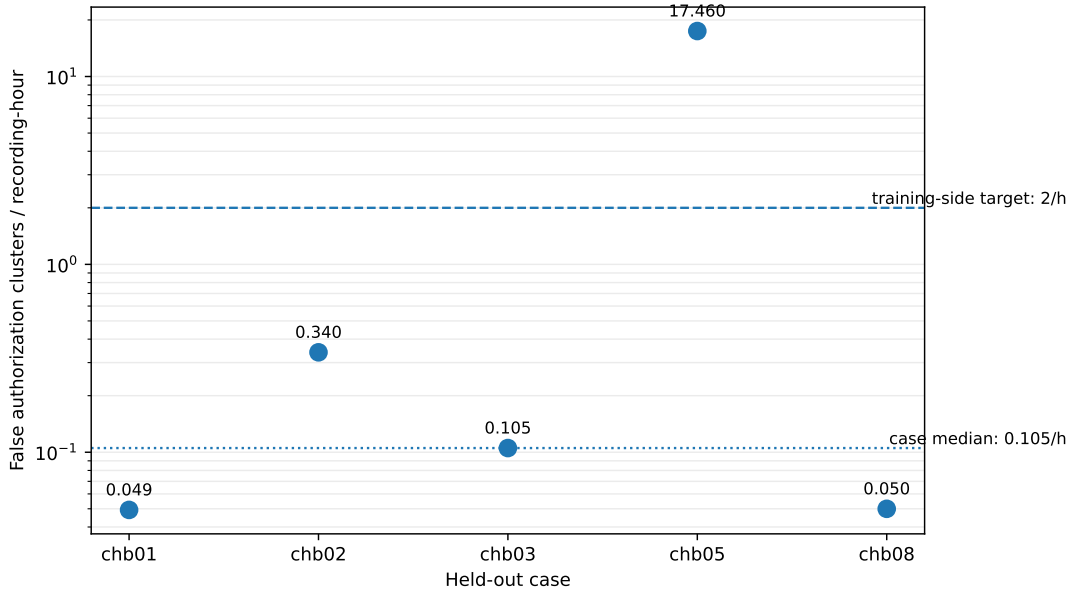


Figure 3: False authorization-cluster burden by held-out case over the full recording inventory. The dashed line marks the 2/h calibration-side target and the dotted line marks the case median (0.105/h). The logarithmic axis makes the pronounced chb05 deviation visible without compressing the other four cases.

3.4 Factorial policy decomposition

On the same calibrated scores and fold-specific thresholds, calibrated thresholding, persistence, attribution overlap alone, and both constraints produced 6.044 ± 12.620 , 3.657 ± 7.875 , 5.791 ± 12.116 , and 3.601 ± 7.748 false clusters/h, respectively (Table 7; Figure 4). Adding persistence reduced the paired case mean by 2.386/h, whereas adding attribution overlap after persistence reduced it by 0.056/h and removed 11 false clusters, all in chb05. The attribution-overlap gate therefore had a measurable but comparatively small incremental effect at $\gamma = 0.4$.

Table 7: Common-score factorial analysis over the full held-out recording inventory. Event categories use window-end decision availability and sum to 27 seizures.

Constraint	False clusters/h	False clusters	Pre / post / missed
Calibrated threshold only	6.044 ± 12.620	1166	3 / 17 / 7
+ persistence	3.657 ± 7.875	711	2 / 18 / 7
+ attribution overlap	5.791 ± 12.116	1117	3 / 17 / 7
+ both	3.601 ± 7.748	700	2 / 18 / 7

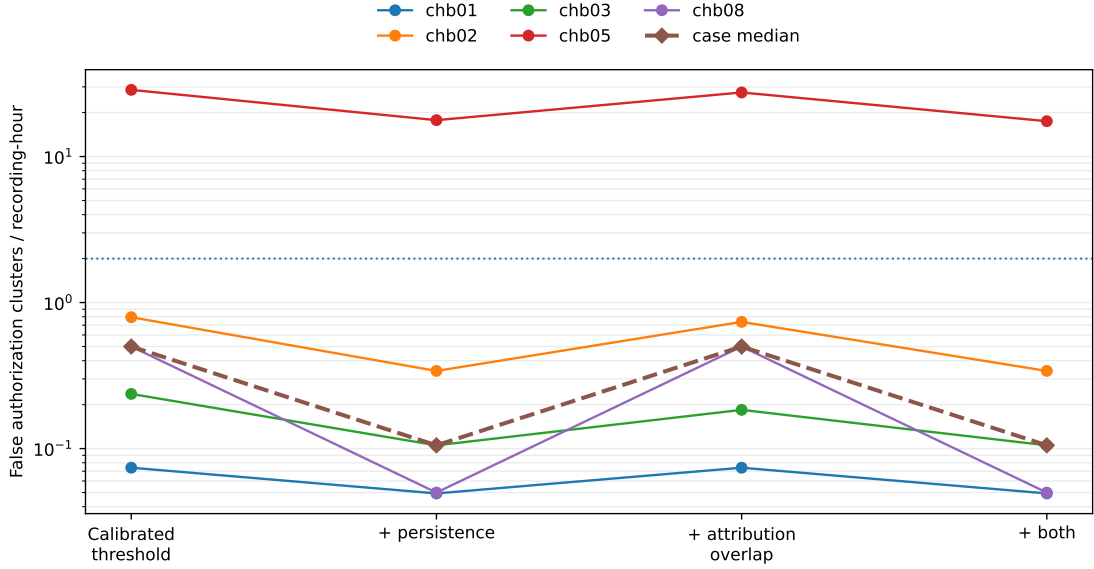


Figure 4: Case-level factorial decomposition of false authorization-cluster burden. All configurations within a case share the same calibrated score and train-selected probability threshold; only persistence and attribution-overlap constraints are toggled. The dotted horizontal line marks 2 false clusters/h and the dashed diamond trace shows the case median.

3.5 Event timing and temporal-rule benchmark

At window-end decision availability, the train-selected persistence-plus-overlap policy authorized 2/27 seizures before onset, 18/27 only at or after onset, and missed 7/27 (Figure 5A). Broad pooled coverage was therefore 20/27 (74.1%), but pre-onset coverage was only 7.4% of all seizures and 10.0% of covered seizures. Median latency among covered events was 16 s (interquartile range 8.5–28 s). The result is consequently more consistent with seizure-responsive authorization than reliable advance seizure prediction.

Under a common 300 s refractory rule and training-only selection objective, persistence, persistence plus overlap, majority-of-three voting, and firing power occupied similar burden-coverage ranges (Table 8; Figure 5B). Persistence plus overlap produced 0.611 ± 1.140 false clusters/h with 17/27 events covered; majority voting produced 0.614 ± 1.193 /h with 16/27 covered; firing power produced 0.517 ± 1.007 /h with 15/27 covered. None of these refractory benchmark policies authorized a seizure before onset. These component-level comparisons therefore do not establish superiority of the attribution-overlap policy.

Table 8: Common-score temporal-rule comparison with 300 s refractory suppression applied to every family. Support fraction is the proportion of recorded time covered by selected EEG-window supports, not VNS dose.

Policy	False clusters/h	Covered events	Support fraction
Threshold	0.786 ± 1.625	16/27	0.10%
Persistence	0.611 ± 1.140	17/27	0.08%
Persistence + overlap	0.611 ± 1.140	17/27	0.08%
Majority of three	0.614 ± 1.193	16/27	0.08%
Firing power	0.517 ± 1.007	15/27	0.07%

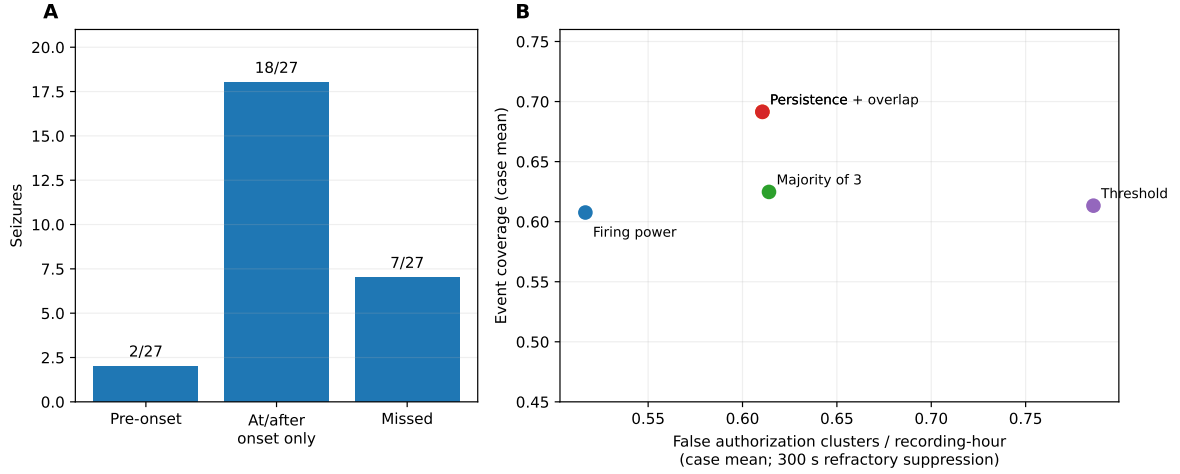


Figure 5: Operational outcomes over the full recording inventory. (A) Seizure categories for the train-selected persistence-plus-overlap policy at window-end decision availability. (B) Burden-coverage positions of adapted temporal-rule families under a common 300 s refractory rule. Points in panel B are equal-weight case means; comparator mechanisms were selected on calibration data only.

3.6 Attribution-overlap sensitivity to temporal separation

The positive-minus-negative attribution-overlap difference remained positive at the first non-overlapping comparison (4 s separation): mean 0.093, exploratory 95% case-bootstrap interval 0.064–0.123 (Figure 6). The mean difference was 0.075 at 6 s, 0.055 at 10 s, and 0.033 at 20 s. At 20 s, one case (chb03) had a negative difference. Thus, the association is not explained solely by the 50% sample overlap of adjacent windows, but it attenuates with temporal separation and does not constitute independent validation of clinical explanation quality.

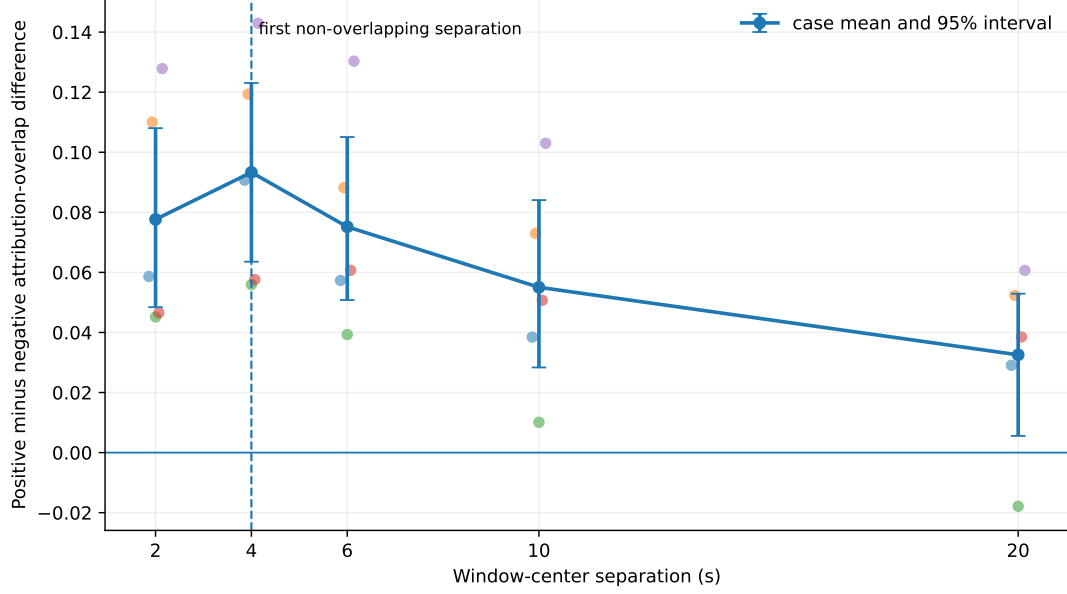


Figure 6: Attribution-overlap difference between positive-label and negative-label authorized windows as a function of temporal separation. Small points show individual held-out cases; the connected points and error bars show the case mean and exploratory 95% case-bootstrap interval. Four seconds is the first separation at which the paired 4 s windows share no raw EEG samples.

3.7 Decision-time sensitivity and chb05 burden distribution

Reclassifying false-cluster truth at window-end decision availability left the principal persistence-plus-overlap endpoint unchanged in every held-out case: the case counts remained 2, 12, 4, 681, and 1 for chb01, chb02, chb03, chb05, and chb08, respectively. The pooled count therefore remained 700 false clusters (4.050/h), the equal-weight case mean remained 3.601/h, and the case median remained 0.105/h. Across the broader factorial set, only one single-window chb05 cluster changed classification under calibrated thresholding alone and attribution overlap alone; neither persistence-based configuration changed. Thus, the principal burden result is insensitive to center-versus-window-end truth assignment.

The chb05 burden was also distributed across recordings rather than being attributable to one or two EDFs. False clusters occurred in 32 of 39 chb05 recordings. The largest contributor, chb05_27.edf, accounted for 86/681 (12.6%) false clusters; the three largest recordings accounted for 33.5%, and the five largest for 48.3%. The 14 original analysis-inventory EDFs contributed 66 false clusters over 14.00 h (4.71/h), whereas the 25 additional evaluation-only EDFs contributed 615 over 25.00 h (24.6/h), with false clusters in 22/25 of those added recordings. A descriptive score comparison likewise showed a heavier upper tail among operationally negative windows in the added exposure: 7.06% exceeded the frozen chb05 threshold $\theta = 0.48$, compared with 1.63% in the original held-out inventory and 0.14% in cross-fitted calibration negatives. These diagnostics support a distributed case-level generalization failure under expanded recording exposure rather than an isolated single-recording anomaly.

4 Discussion

The principal finding of this study was that a policy satisfying its burden constraint during calibration could still show pronounced case-specific failure when evaluated over complete held-out recording exposure. Across the complete 175-EDF evaluation, four of the five held-out cases

remained between 0.049 and 0.340 false authorization clusters per analyzed recording-hour, whereas chb05 reached 17.460/h. The targeted EDF-level diagnostic showed that this chb05 burden was distributed across 32 of 39 recordings rather than driven by a single anomalous file, and it increased particularly in the additional evaluation-only exposure. This divergence shows that policy-level generalization cannot be inferred from discrimination or calibration-side operating constraints alone and makes complete-recording exposure an important component of deployment-oriented evaluation. This result is important because seizure-detection and seizure-prediction systems are commonly evaluated using aggregate sensitivity and false-alarm metrics, even though alarm burden can depend strongly on patient-specific signal characteristics and on the temporal rules that convert classifier scores into actionable events [1, 2, 9]. Our results extend that concern from classifier performance to the policy layer itself.

The common-score factorial analysis identifies temporal persistence as the principal burden-reducing component among the two policy constraints examined. Persistence reduced the case-mean false-cluster rate by 2.386/h relative to calibrated thresholding, whereas attribution overlap reduced burden only modestly after persistence. This distinction matters because an observable attribution property need not translate into a large operational benefit. Calibration, temporal regularization, and attribution gating should therefore be evaluated as separate policy components rather than inferred from one composite comparison.

Attribution overlap nevertheless showed a persistent association with the operational label beyond the adjacent-window sample overlap. The positive-minus-negative difference remained positive on average at 4–20 s separations, although it attenuated with lag and became negative in one case at 20 s. These findings support attribution overlap as a reproducible model-behavior descriptor, not as evidence that the resulting explanations are clinically faithful. The latter would require an independent explanatory reference, such as expert-rated physiological relevance, which is absent here[12, 14].

The temporal-rule benchmark further limits claims of policy-specific superiority. Majority voting, persistence, persistence plus overlap, and firing power occupied similar burden–coverage ranges under a common refractory rule. This is consistent with the broader literature showing that post-processing choices can materially change alarm behavior[1, 2]. The unsuppressed firing-power analysis, reported in the supplement, also illustrates why cluster frequency must be interpreted together with selected-support duration: sustained authorization can collapse into a small number of long clusters without representing low intervention burden.

Event timing further distinguishes the authorization task from advance prediction. Only 2 of 27 seizures were authorized before onset, whereas 18 were first authorized at or after onset, with a median covered-event latency of 16 s. This pattern is consistent with the operational target, which combines preictal and ictal windows and is therefore evaluated as seizure-responsive authorization, with pre-onset performance reported separately. Reclassification of false-cluster truth at window-end availability left the principal burden endpoint unchanged, showing that the main burden result is not an artifact of the center-versus-end timing convention. No VNS was delivered, and post-onset authorization cannot be interpreted as evidence of seizure termination or therapeutic benefit.

5 Limitations

The study contains only five CHB-MIT cases, selected as a convenience sample, and does not provide an independent patient cohort. The original 70-EDF analysis inventory was not defined by a documented prospective clinical sampling rule and contains all seizure-bearing EDFs from these cases, whereas the 105 additional evaluation-only EDFs contain no annotated seizures. This design tests expanded held-out recording exposure without retuning but does not establish population representativeness. The strong influence of chb05, despite being distributed across many of its recordings, further demonstrates why case-level results must accompany averages.

Labels use a fixed, record-local preictal-plus-ictal convention rather than a direct measure of physiological preictal state. Preictal intervals were not propagated across EDF boundaries, so recordings without seizure annotations are operationally negative under this convention rather than guaranteed physiologically interictal. The 2/h calibration target is an analytical operating constraint, not a validated clinical utility limit. Window-end timing omits unmeasured computation, communication, and stimulation delays.

Attribution overlap has no independent clinical explanatory ground truth. The comparator experiments adapt temporal rule components on common scores rather than reproduce complete external systems. All analyses are retrospective and evaluate simulated authorizations only; they do not address stimulation efficacy, adverse effects, seizure termination, battery use, or long-term clinical response.

6 Conclusion

EEG-guided adaptive-neuromodulation simulations should be evaluated at the level of the authorization policy, not by classifier discrimination alone. Across the full recording inventory of five CHB-MIT cases, authorization burden remained low in four held-out cases but increased sharply and across many recordings in one, showing that a training-side burden constraint does not provide a subject-independent deployment guarantee. Temporal persistence accounted for most of the burden reduction, attribution overlap added a smaller secondary constraint, and most covered seizures were authorized only after onset. The resulting framework provides a reproducible basis for evaluating calibration, temporal policy, attribution behavior, recording exposure, burden, and event timing before predictive or therapeutic device validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and Code Availability

CHB-MIT EEG is publicly available through PhysioNet[8]; raw EEG recordings are not redistributed. Analysis code, derived inputs, fitted model artifacts, parameter searches, event/cluster outputs, tests, and figure-generation scripts are maintained in the [project repository](#). The versioned archival DOI corresponding to this manuscript release is reported in the repository release metadata.

Ethics Statement

This study used de-identified public EEG data and did not involve direct interaction with human participants.

CRedit Authorship Contribution Statement

Suzana Cescon: Methodology, Software, Formal analysis, Validation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

Antonio Pereira: Conceptualization, Supervision, Methodology, Clinical and ethical framing, Project administration, Writing – original draft, Writing – review and editing, Corresponding author.

A Supplementary Methods and Policy Tables

A.1 Detailed fold partitions

Table 9: Nested partition composition for within-subject chronological folds. “Sz rec.” is the number of seizure-containing EDFs.

Fold	Core model				Calibration				Test			
	EDF	h	Pos. win.	Sz rec.	EDF	h	Pos. win.	Sz rec.	EDF	h	Pos. win.	Sz rec.
chb01	8	8.00	506	3	4	4.00	372	2	8	6.55	398	2
chb02	3	2.27	106	1	2	2.00	191	1	4	4.00	155	1
chb03	6	6.00	545	3	2	2.00	177	1	5	5.00	534	3
chb05	5	5.00	208	1	3	3.00	206	1	6	6.00	617	3
chb08	6	6.00	700	3	2	2.00	230	1	6	6.00	282	1

Table 10: Nested partition composition for held-out-subject folds. “Sz rec.” is the number of seizure-containing EDFs.

Held out	Core model				Calibration				Test			
	EDF	h	Pos. win.	Sz rec.	EDF	h	Pos. win.	Sz rec.	EDF	h	Pos. win.	Sz rec.
chb01	35	34.27	2,945	15	15	15.00	1,006	5	20	18.55	1,276	7
chb02	43	42.74	3,723	19	18	16.81	1,052	5	9	8.27	452	3
chb03	40	39.01	3,124	16	17	15.81	847	4	13	13.00	1,256	7
chb05	39	38.01	3,198	17	17	15.81	998	5	14	14.00	1,031	5
chb08	39	38.01	3,090	17	17	15.81	925	5	14	14.00	1,212	5

A.2 Nominal policy-configuration comparison

Table 11: Prespecified LightGBM policy configurations at $\theta = 0.5$. These rows are descriptive configurations rather than an isolating ablation because probability inputs and rule components differ across rows. Values are mean $\pm$ SD across five folds.

Validation	Configuration	Auth. clusters/h	False clusters/h	Coverage	Pos. auth. rate
Within-subject	Fixed threshold	44.11 ± 8.84	40.37 ± 10.51	1.00 ± 0.00	0.216
	Temporal persistence	18.06 ± 5.39	15.67 ± 5.43	1.00 ± 0.00	0.344
	Attribution-overlap gate	41.50 ± 6.80	37.87 ± 8.52	1.00 ± 0.00	0.219
	Calibration-aware gate	4.82 ± 2.41	3.53 ± 1.96	1.00 ± 0.00	0.604
	Full governed	1.67 ± 0.92	0.95 ± 0.80	0.93 ± 0.15	0.740
Held-out	Fixed threshold	88.38 ± 30.05	84.44 ± 29.57	1.00 ± 0.00	0.093
	Temporal persistence	49.14 ± 16.77	46.01 ± 16.69	1.00 ± 0.00	0.135
	Attribution-overlap gate	79.18 ± 26.55	75.23 ± 26.26	1.00 ± 0.00	0.101
	Calibration-aware gate	3.31 ± 4.67	2.70 ± 4.60	0.76 ± 0.29	0.637
	Full governed	1.72 ± 2.10	1.17 ± 2.05	0.76 ± 0.29	0.703

A.3 Full nominal event mapping

Table 12: Development-exposure, center-referenced nominal event mapping for LightGBM configurations. Pre-onset counts are shown among all events; post-onset-only and missed counts are derived from pooled event mappings. Lead time is the mean $\pm$ SD of fold-specific median lead times among covered events.

Validation	Configuration	Covered	Pre-onset	Post-onset only	Missed	Lead (s)
Within-subject	Fixed threshold	10/10	10	0	0	259.5 ± 49.9
	Temporal persistence	10/10	7	3	0	155.3 ± 149.4
	Attribution-overlap gate	10/10	9	1	0	245.3 ± 79.0
	Calibration-aware gate	10/10	4	6	0	72.1 ± 105.3
	Full governed	9/10	2	7	1	26.1 ± 92.5
Held-out	Fixed threshold	27/27	24	3	0	235.8 ± 60.7
	Temporal persistence	27/27	22	5	0	184.0 ± 113.5
	Attribution-overlap gate	27/27	23	4	0	234.6 ± 61.3
	Calibration-aware gate	19/27	3	16	8	7.7 ± 57.5
	Full governed	19/27	1	18	8	-17.4 ± 10.8

A.4 Cooldown sensitivity

Cooldown changes cluster consolidation but not the underlying authorization-window mask. It therefore preserves each policy’s own event coverage; the fixed and full policies in Table 13 are not matched at a common coverage level.

Table 13: Held-out-subject cooldown sensitivity for the nominal LightGBM fixed-threshold and full configurations. Values are mean $\pm$ standard deviation across five folds.

Policy	Cooldown (s)	Auth. clusters/h	False clusters/h	Median duration (s)	Coverage
Fixed threshold	0	88.38 ± 30.05	84.44 ± 29.57	4.8 ± 1.1	1.00 ± 0.00
	30	20.42 ± 4.10	19.49 ± 4.10	22.0 ± 7.9	1.00 ± 0.00
	60	10.42 ± 1.36	9.79 ± 1.39	51.6 ± 15.5	1.00 ± 0.00
	120	5.11 ± 0.92	4.67 ± 0.82	184.4 ± 94.0	1.00 ± 0.00
Full governed	0	1.72 ± 2.10	1.17 ± 2.05	5.2 ± 1.1	0.76 ± 0.29
	30	1.10 ± 1.20	0.76 ± 1.15	7.8 ± 5.8	0.76 ± 0.29
	60	0.97 ± 1.01	0.64 ± 0.95	8.4 ± 5.7	0.76 ± 0.29
	120	0.86 ± 0.82	0.56 ± 0.77	13.2 ± 9.2	0.76 ± 0.29

B Case-level event timing

Table 14: Train-selected policy at window-end decision time. Counts are among all events in each case. Latency summaries include covered events only.

Case	Events	Pre-onset	Post-onset only	Missed	Covered	Median latency (s)
chb01	7	0	6	1	6	35.0
chb02	3	0	3	0	3	12.0
chb03	7	0	2	5	2	28.0
chb05	5	2	2	1	4	-86.5
chb08	5	0	5	0	5	18.0
Pooled	27	2	18	7	20	16.0

The pooled latency interquartile range is 8.5–28 s. The pooled lead median is -16 s; the mean $\pm$ SD of the five case-specific median leads is -1.3 ± 49.9 s. The latter is not a pooled

seizure-level median. A negative case median latency can coexist with substantial post-onset-only counts when a case has both early and late authorizations. Pre-onset coverage is 2/27 among all events and 2/20 among covered events. Full per-event timestamps and the center-to-end comparison are provided in `results/event_timing_by_event.csv`.

C Reproducibility Materials

Table 15: Reproducibility materials supplied with the analysis package.

Artifact group	Included materials
Dataset manifest	EDF file list, configured cases, recording metadata, and CHB-MIT directory structure.
Analysis configuration	Window length, overlap, preictal horizon, feature families, model settings, calibration settings, attribution-overlap parameters, thresholds, and cooldown settings.
Validation files	Held-out-subject and within-subject EDF-level splits; nested core/calibration/test composition; train-test overlap checks; chronological-order checks.
Model outputs	Raw probabilities, calibrated probabilities, labels, case IDs, EDF identifiers, window timestamps, and performance summaries.
Calibration and attribution	Reliability data, Brier score, ECE, calibration slope/intercept, cross-fitted calibrator selection, top-feature sets, and temporal attribution-overlap analyses.
Policy and event outputs	Authorization windows and clusters, false-cluster burden, train-only operating-point selection, factorial policy results, cooldown sensitivity, event categories, and lead-time summaries.
Reproduction files	Figure-generation scripts, manuscript-generation scripts, package-lock file, exported environment file, and exact software versions.

D Supplementary Operating-Point Sensitivity

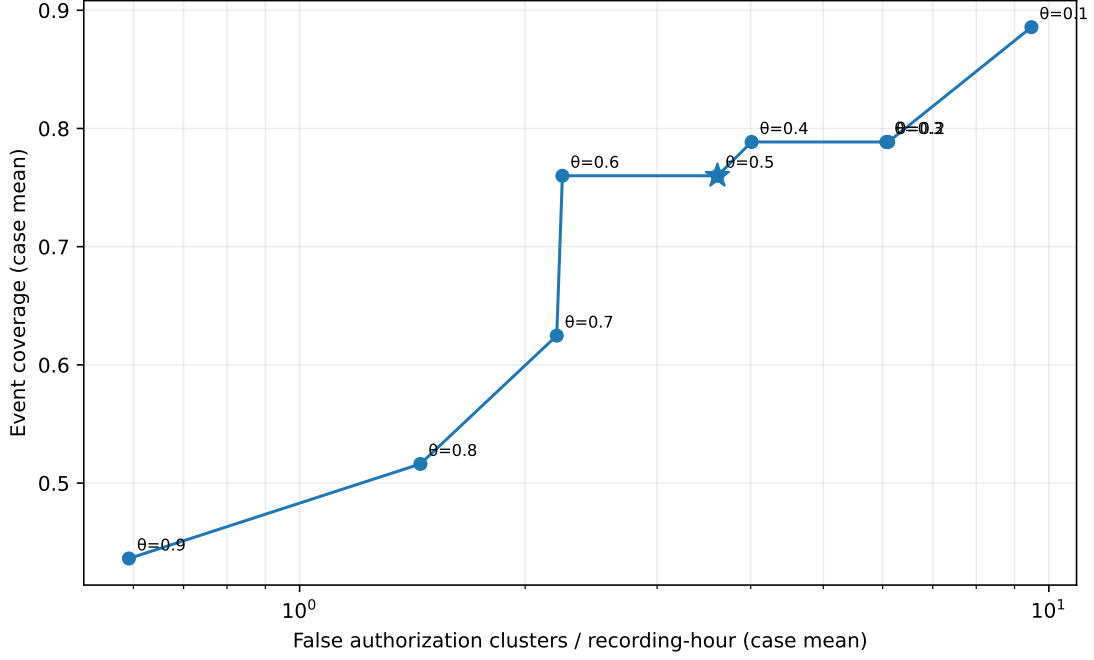


Figure S1: Probability-threshold sensitivity with the attribution-overlap threshold fixed at $\gamma = 0.4$. The star marks the prespecified $\theta = 0.5$ configuration; it is not a test-selected optimum. Error bars at the prespecified point are exploratory 95% case-bootstrap intervals. Full case-level sweep values are provided with the reproducibility package.

E Paired comparator uncertainty

Table 16: Paired case-level false-cluster-rate differences relative to persistence plus attribution overlap under the common 300 s refractory rule on the complete recording inventory. Positive values indicate a higher false-cluster rate than the persistence-plus-overlap baseline. Intervals and sign-flip probabilities are descriptive because there are only five cases.

Comparison	Mean difference (/h)	95% case-bootstrap interval	Sign-flip p	Cases lower/equal/higher
Threshold	0.175	$[-0.126, 0.631]$	0.750	1/2/2
Persistence	0.000	$[0.000, 0.000]$	1.000	0/5/0
Majority of three	0.003	$[-0.051, 0.062]$	1.000	1/3/1
Firing power	-0.094	$[-0.267, 0.083]$	0.313	4/0/1

F Targeted robustness diagnostics

Table 17: Decision-time sensitivity of the principal train-selected persistence-plus-overlap policy. False clusters are classified either from the original center-based operational labels or from operational labels evaluated at window-end decision availability.

Case	Center-label false clusters	End-time false clusters	Evaluated h	End-time rate (/h)
chb01	2	2	40.55	0.049
chb02	12	12	35.27	0.340
chb03	4	4	38.00	0.105
chb05	681	681	39.00	17.460
chb08	1	1	20.01	0.050
Pooled	700	700	172.83	4.050

Table 18: Distribution of chb05 false-cluster burden across the original analysis inventory and the additional evaluation-only exposure for the principal train-selected policy.

Exposure	EDFs	EDFs with false clusters	Evaluated h	False clusters	False clusters/h
Original analysis inventory	14	10	14.00	66	4.71
Additional evaluation only	25	22	25.00	615	24.60
Combined	39	32	39.00	681	17.46

For chb05, the largest single EDF contributed 86/681 (12.6%) false clusters, the three largest contributed 33.5%, and the five largest contributed 48.3%. Among operationally negative windows, the fraction of calibrated scores exceeding the frozen threshold $\theta = 0.48$ was 0.14% in cross-fitted calibration predictions, 1.63% in the original held-out analysis inventory, and 7.06% in the additional evaluation-only recordings. Full EDF-level values and score quantiles are provided in `results/chb05_per_edf_diagnostic.csv` and `results/chb05_score_shift_summary.csv`.

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